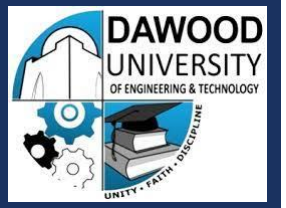
LECTURE – ENCAPSULATION
OOP (JAVA)



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AGENDA



- INTRODUCTION TO ENCAPSULATION
- ADVANTAGES OF ENCAPSULATION
- EXAMPLE
- ACCESSOR AND MUTATORS

ENCAPSULATION

- **Encapsulation in Java** is a *process of wrapping code and data together into a single unit*, for example, a capsule which is mixed of several medicines.
- We can create a fully encapsulated class in Java by making all the data members of the class private.
- Now we can use setter and getter methods to set and get the data in it.



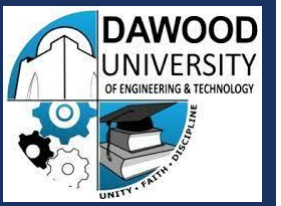
Capsule

ADVANTAGES OF ENCAPSULATION



- By providing only a setter or getter method, you can make the class **read-only or write-only**. In other words, you can skip the getter or setter methods.
- It provides you the **control over the data**.
- Suppose you want to set the value of id which should be greater than 100 only, you can write the logic inside the setter method.
- You can write the logic not to store the negative numbers in the setter methods.
- It is a way to achieve **data hiding** in Java because other class will not be able to access the data through the private data members.
- The standard IDE's are providing the facility to generate the getters and setters. So, it is **easy and fast to create an encapsulated class** in Java.

READ ONLY CLASS




```
//A Java class which has only getter methods.
public class Student{
//private data member
private String college="AKG";
//getter method for college
public String getCollege(){
return college;
}
}
```

WRITE ONLY CLASS

```
//A Java class which has only sette
r methods.
public class Student{
//private data member
private String college;
//getter method for college
public void setCollege(String colle
ge){
this.college=college;
}
}
```



THANK YOU